Patient Health Record Consent Management System

TEAM : Nishitha S , Narayanan Arun , Shashank M P , Nishanth N

Lab 3: Component Modelling & Architectural Pattern Selection

Architecture Selection: Layered Architecture

Justification

We selected Layered Architecture for the Patient Health Record Consent Management System because it separates user interaction, consent management, access control, and data storage into distinct components with clearly defined responsibilities.

Reason 1 — Secure Medical Record Access

The system must ensure that verified doctors can access only the medical records explicitly permitted by patients. The Consent Management Service and Health Record Gateway work together to validate consent status, record scope, and expiry before releasing medical information. This separation makes access-control logic easier to maintain and test.

Reason 2 — Maintainability and Modularity

The system includes several independent responsibilities, including patient authentication, consent granting and revocation, record retrieval, and audit logging. Organising these responsibilities into separate components allows individual modules to be updated without requiring major changes to the entire system.

Security Advantage

The Identity & Access Verification Service validates user identity and role, while the Health Record Gateway enforces consent restrictions before releasing records. The Audit Logging Service records access attempts and consent changes in an append-only trail, improving accountability and traceability.

Performance Benefit

The Health Record Gateway can perform focused authorization checks before retrieving medical records, avoiding unnecessary data retrieval for unauthorized requests. This supports the system's requirement to complete at least 95% of authorization requests within two seconds under the specified test load.